

EASTERN SIERRA, CA, USA

WMO: 724800

Lat: 37.371N

Lon: 118.358W

Elev: 1250

StdP: 87.18

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23157

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-8.8	-7.0	-21.4	0.6	6.6	-19.3	0.8	7.9	12.7	7.5	11.6	7.2	2.5	320	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	38.6	16.7	37.4	16.3	36.2	15.8	18.8	33.9	17.9	33.0	17.1	32.4	4.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.1	11.7	22.6	12.1	10.3	24.1	10.3	9.1	23.9	58.1	33.6	55.0	32.8	52.3	32.4	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	10.0	8.7	DB	-12.7	40.8	1.7	1.0	-13.9	41.5	-14.9	42.1	-15.9	42.6	-17.1	43.4	(4)
(5)			WB	-13.7	19.8	1.8	1.2	-15.0	20.6	-16.1	21.3	-17.1	22.0	-18.4	22.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.9	4.3	5.7	9.6	12.6	17.2	22.5	25.6	24.3	20.1	13.9	7.5	3.4	(6)	
	DBStd	8.35	3.50	3.68	3.69	3.95	3.86	3.49	2.26	2.08	3.01	3.58	3.88	3.42	(7)	
	HDD10.0	683	177	125	54	21	2	0	0	0	0	10	91	204	(8)	
	HDD18.3	2257	434	353	272	174	68	9	0	0	18	143	324	462	(9)	
	CDD10.0	2118	1	6	40	98	224	374	484	442	302	130	17	1	(10)	
	CDD18.3	650	0	0	0	2	32	133	226	184	69	4	0	0	(11)	
	CDH23.3	13181	1	2	39	217	936	2625	3961	3402	1658	327	12	0	(12)	
(13)	CDH26.7	7436	0	0	3	45	385	1525	2500	2072	822	83	1	0	(13)	
(14)	Wind	WSAvg	3.5	3.1	3.6	4.0	4.4	3.9	3.7	3.5	3.4	3.4	3.1	3.1	(14)	
(15) Precipitation	PrecAvg	138	24	25	11	6	6	4	4	4	5	3	14	18	(15)	
	PrecMax	434	227	153	75	41	33	33	18	16	30	20	66	147	(16)	
	PrecMin	46	0	0	0	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	81	39	38	15	8	9	8	5	5	7	5	18	27	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.7	23.1	26.8	30.3	35.1	39.4	40.6	39.2	36.1	31.4	25.0	19.6	(19)	
		MCWB	7.9	8.5	10.5	12.3	14.7	16.2	17.5	17.3	15.9	13.5	10.2	7.5	(20)	
	2%	DB	18.0	20.4	24.3	27.9	32.7	37.5	39.1	37.8	34.6	29.0	22.6	16.9	(21)	
		MCWB	6.7	7.5	9.3	11.1	13.5	15.5	17.1	16.5	15.2	12.4	9.3	6.1	(22)	
	5%	DB	15.8	18.1	22.3	26.0	30.9	36.0	37.8	36.7	33.3	27.0	20.2	14.7	(23)	
		MCWB	5.9	6.7	8.6	10.4	12.9	15.2	16.6	16.0	14.7	11.4	8.2	5.2	(24)	
	10%	DB	13.2	15.4	20.3	23.8	28.8	34.5	36.6	35.5	31.7	24.9	17.7	12.1	(25)	
		MCWB	4.9	5.6	7.7	9.4	12.2	14.8	16.2	15.6	14.0	10.4	6.9	4.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	9.6	11.2	13.4	15.8	18.6	20.2	19.9	17.9	15.1	11.2	9.0	(27)	
		MCDB	18.2	19.8	24.6	28.0	32.7	36.1	35.7	35.6	32.0	25.0	22.8	16.9	(28)	
	2%	WB	7.9	8.3	10.0	11.7	14.4	16.9	19.1	18.4	16.9	13.4	9.9	7.3	(29)	
		MCDB	15.0	18.7	22.8	26.3	30.4	34.7	34.0	33.7	31.1	26.2	20.9	14.5	(30)	
	5%	WB	6.7	7.2	9.0	10.7	13.5	16.1	18.3	17.5	15.8	12.1	8.7	5.8	(31)	
		MCDB	14.1	16.6	21.2	24.7	28.9	33.8	33.0	32.9	30.1	25.2	19.0	13.1	(32)	
	10%	WB	5.4	6.0	8.0	9.8	12.5	15.2	17.5	16.7	14.7	11.0	7.5	4.6	(33)	
		MCDB	12.8	14.3	19.5	23.1	27.3	32.6	32.3	32.0	29.4	23.4	17.1	11.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	16.3	16.4	18.3	18.5	19.4	21.4	21.8	22.5	21.9	20.3	18.8	16.4	(35)	
		MCDBR	20.7	21.4	21.7	22.2	22.5	23.4	23.3	23.6	23.6	23.3	22.0	20.5	(36)	
	5% WB	MCWBR	11.5	11.0	10.3	9.6	8.6	8.3	8.2	8.6	9.6	10.6	11.5	11.8	(37)	
		MCWBR	18.1	19.2	20.6	21.1	21.5	22.2	20.4	21.2	22.0	21.2	20.7	18.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.262	0.276	0.287	0.318	0.333	0.330	0.340	0.328	0.305	0.289	0.265	0.261	(40)	
		taud	2.473	2.446	2.443	2.398	2.415	2.437	2.470	2.478	2.545	2.567	2.576	2.513	(41)	
	Ebn at Noon		941	966	983	962	946	945	933	940	951	941	935	921	(42)	
		Edh at Noon	80	94	104	116	116	114	109	106	93	83	72	72	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.70	3.61	4.95	6.35	7.22	8.31	7.74	7.27	6.19	4.56	3.19	2.41	(44)	
	RadStd		0.23	0.34	0.34	0.34	0.56	0.53	0.47	0.39	0.24	0.31	0.18	0.24	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.38	N/A	-1.46	N/A	N/A	N/A	N/A	N/A	+122	+81	(46)
(47)	Regional (14 neighbors)	+0.54	N/A	N/A	+0.40	N/A	N/A	-45	N/A	+152	+92	(47)

Nomenclature: See separate page